

Finnhouses AI Platform Architecture Blueprint

Executive Reference

Version: Final

Status: Architecture Baseline

Scope: Assessment → Diagnosis → Strategic Options → Migration → Governance → Executive Decision

Freeze Status: Architecture Freeze Maintained

Next Document: Finnhouses Intelligence Governance Implementation Plan v1.0

Executive Brief

Executive Overview

Vision:

AI-native, Human-Centered Real Estate Intelligence Operating System.

Current Position:

Finnhouses has already passed the “build an AI platform” phase. It has AI agents, automation runtime, a Hub architecture, a data layer, human feedback mechanisms, encoded business rules, and real operational workflows generating revenue today.

Main Finding:

The platform is not lacking AI capability. It is lacking the operating model that allows AI capability to compound.

Strategic Direction:

Add a Governance and Intelligence Layer on top of what exists. Do not replace components, do not add agent count for its own sake, do not adopt enterprise-scale governance prematurely.

Current Architecture Assessment

Business	
Applications	(Next.js / Hub v1 / Hub v2 / n8n)
Automation	(34 Hub routes, ~16 n8n workflows)
AI Agents	(5: Content, Quality Gate, QC Vision, Market Intel, CRM Parser)
Data	(26 Supabase tables, RLS enabled)

Missing at every layer: No Intelligence Governance Layer — no agent registry, no unified decision ledger, no cross-agent memory, no scoped AI access control.

Three Cross-Cutting Findings

1. Fragmentation Pattern

Agents, data, runtime, and governance all fracture along the same silo lines; one root cause appearing four times.

2. Maturity Inversion Pattern

The most architecturally mature capability (Hub v2's AiGateway, HealthMonitor, auth model) exists outside the production boundary. The organization has the solution; it lacks the migration mechanism.

3. Capability > Governance Gap

The platform accumulates data volume, not compounding intelligence.

Strategic Decisions Required

Decision #1 — AI Governance Model:

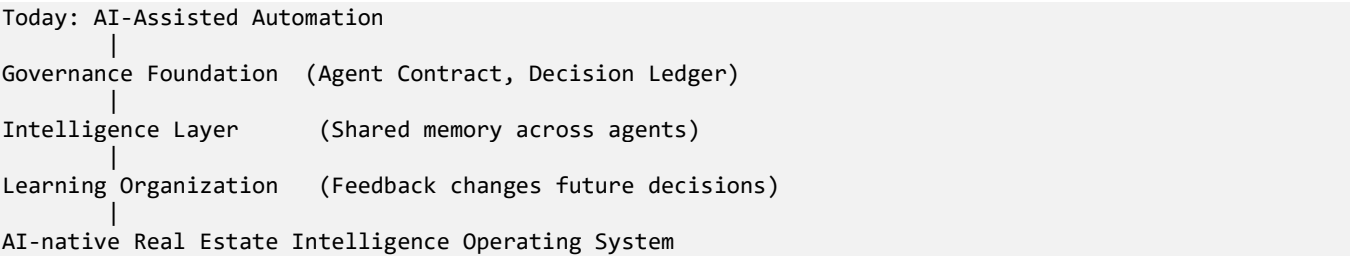
Distributed / Control Plane / Hybrid Intelligence Layer.

Decision #2 — Hub Evolution:

Big Bang / Partial Migration / Extract Gradually.

These are the only two decisions requiring Archi's strategic judgment. Everything else (credential isolation, monitoring, deployment confidence, incident response) is a no-regret action proceeding independently.

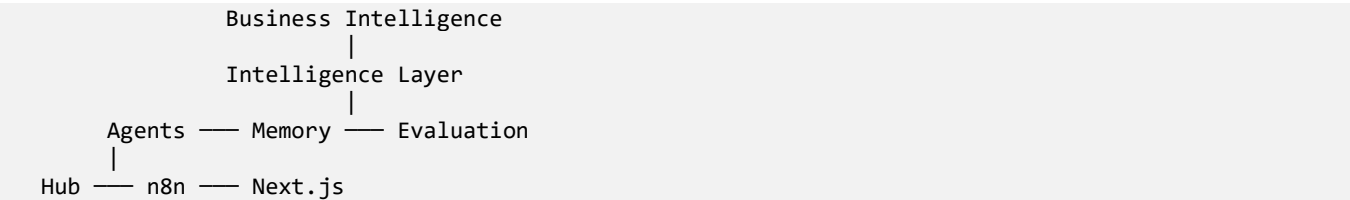
Recommended Evolution Path



Immediate Action Plan (0–90 days, no-regret)

- Credential isolation — per-service Supabase keys; before: 1 leak = platform compromise, after: 1 leak = 1 service impact.
- Operational monitoring — health parity for Hub v1, queue-stuck detection, workflow failure alerting.
- Deployment confidence — smoke tests beyond git-SHA verification.
- Incident response runbook — disable → rotate → update services → verify → review.
- Agent Contract prototype — pilot on AG-4.2 (Quality Gate): existing feedback, existing logging, clear rule, measurable outcome.

Future State Architecture



Final Executive Decision

1. Will AI be built as a Feature, or as an Operating System?
2. Should Governance precede Scale, or Scale precede Governance?
3. Is Hub v2 the Destination, or a Capability Source to extract from?

Final Summary

Cross-Cutting Patterns

Pattern 1 — Fragmentation Pattern

Agents, Data, Runtime, and Governance all fracture along the same lines, independently — one structural tendency appearing four times.

Pattern 2 — Maturity Inversion Pattern

The most mature capability consistently exists outside the active production boundary. The organization is not short on capability; it is short on a mechanism to carry capability into production.

Pattern 3 — Capability > Governance Gap

Finnhouses is not lacking AI capability. It is lacking the operating model that allows AI capability to compound.

Architecture Maturity / Readiness Assessment

Dimension	Current Level	Evidence
AI Capability Existence	Present (5 agents)	Ch4
Human-in-the-loop Audit	Present, 3 of 5 agents	Ch4, Ch7
Decision Logging	Present, fragmented (3 schemas)	Ch5 Major Finding 5.1
Cross-Agent Intelligence Reuse	Absent	Ch7 Major Finding 7.1
Learning Loop	Absent	Ch7
AI Governance Boundary	Absent	Ch8 Major Finding 8.1
Proactive Operations	Absent	Ch9 Major Finding 9.1

Overall platform maturity: L1–L2 of L5 (Automated Operations with Audit Capability).

This is a factual position statement, not a criticism — the platform functions and generates real business value today. The gap is between current state and the stated Vision, not between current state and a functioning business.

Top Risks (ranked by blast radius)

- Privileged credential exposure (Major Finding 8.1) — highest blast radius identified anywhere in the Blueprint.
- Reactive operations model (Major Finding 9.1) — 13 of 15 documented incidents discovered incidentally.
- Governance-free automation layer (AO-6.5) — n8n hosts 3 of 5 AI agents including the one enforcing a Critical business rule.
- Hub v1/v2 coexistence drift — indefinite dual-maintenance cost and risk of a rushed cutover.

Top Strategic Opportunities

- The Hybrid Intelligence Layer (Cluster A, Option C) — achievable without disrupting existing team velocity.
- AG-4.2 (Quality Gate) as pilot governance agent — lowest-risk, highest-learning entry point.
- No-regret Track B work — can start immediately and measurably reduces the two highest-ranked risks.

- Hub v2's “extract gradually” path (Option 3) — components may deliver value back-ported into Hub v1 sooner than a full cutover.

Open Questions

- AI Execution Governance Strategy — Distributed / Control Plane / Hybrid (Decision #1).
- Hub v1/v2 Evolution Path — Big Bang / Partial / Extract Gradually (Decision #2).
- Governance role assignment given actual team size (Confidence: Low).
- Governance Maturity Level threshold before adding a 6th AI agent.
- Table-count discrepancy between Ch2 (28) and Ch5 (26) — flagged, not resolved.

Executive Conclusion

Finnhouses has successfully built the foundation of an AI-enabled real estate platform: automation exists, AI agents exist, operational workflows exist. However, the platform has reached the point where additional capability without governance will increase complexity faster than intelligence.

The next evolution is therefore not adding more AI agents or replacing existing components. The next evolution is introducing the Intelligence Governance Layer: Agent Identity, Decision Memory, Evaluation, Access Control, and Operational Discipline.

The strategic objective is to move from AI-assisted automation toward a self-improving Intelligence Operating System.

Top 10 Architecture Decisions

#	Decision	Converges	Urgency
1	AI Execution Governance Strategy (Distributed / Control Plane / Hybrid)	ADR 4.1,5.1,5.3,6.4,7.1,7.2,8.4	Strategic — foundational
2	Hub v1/v2 Evolution Path	ADR 2.1, 6.1, 6.2	Strategic — not blocking Track B
3	Credential Isolation (Blast Radius Reduction)	ADR 8.2 Phase 1	No-regret — now
4	AI Decision & Feedback Data Model	ADR 5.1	Depends on #1
5	Automation Governance Model (n8n)	ADR 6.3	Medium
6	Proactive Operations	ADR 9.1, 9.2	No-regret — now
7	Deployment Confidence	ADR 9.3	No-regret — now
8	Credential Compromise Runbook	ADR 9.4	No-regret — now
9	Business Rule Data Store	ADR 5.4, 4.2	Depends on #1
10	Header/Auth Convention Standardization	ADR 8.1	Low-cost, now

Decisions #3, #6, #7, #8, #10 require no strategic choice — they can proceed immediately. Decisions #1 and #2 are the two genuine strategic forks; #4, #5, #9 are downstream of #1.

Roadmap Summary

Track A: Strategic Migration

S0 Governance Foundation Phase

-> Agent Identity, Decision Ledger, Memory Contract, Access Policy
(Logical Capability Objects, not schema yet)

Gate 1: Intelligence Foundation Ready

-> Registry + Log + Feedback + Evaluation

S1 Intelligence Layer Introduction

Gate 2: Production Governance Ready

-> Identity + Permission + Monitoring + Rollback + Evaluation

S2 Agent Governance Migration

-> AG-4.2 -> AG-4.3 -> AG-4.4 -> AG-4.5 -> AG-4.1

Gate 3: Runtime Evolution Decision

-> Option 1 / 2 / 3 (decided with real operating data)

Track B: Operational Hardening (No-Regret, starts now)

- O1 — Credential Blast Radius Reduction
- O2 — Operational Observability (health parity, alerting)
- O3 — Deployment Confidence (smoke tests, functional verification)
- O4 — Incident Response Runbook (Response side)

Roadmap Guardrails

- Guardrail 1: No new production AI agents before Gate 1.
- Guardrail 2: No agent migration before Gate 2.
- Guardrail 3: No Hub runtime decision before Gate 3 evidence.
- Guardrail 4: No governance bypass for speed — experimental speed belongs in the sandbox layer only.

Migration Risk Register

Risk	Trigger	Mitigation
Governance design delays feature work	Team adds agents faster than governance is built	Sandbox rule (Guardrail 4)
Hub decision paralysis	Waiting for a “perfect” architecture	Extract-capability approach (Option 3) remains available
Over-engineering the Intelligence Layer	Designing schema before usage evidence exists	Logical Capability Objects first, schema deferred
Migration fatigue	Too many parallel changes	Track B (parallel) + Gated Track A (sequential)

Governance Model

Governance Objective

Build Faster → Build → Govern → Learn → Improve. Not a rejection of build speed — Decision Principle 1 and Guardrail 4 both explicitly protect it.

Minimum Agent Contract

Every production AI agent must define: Agent Identity + Purpose + Input Data + Output Type + Decision Impact + Evaluation Method + Owner.

Production entry gate: Agent Registration → Permission Review → Evaluation Dataset → Production Approval.

Change Classification

Level	Examples	Governance Requirement
L1 — Experiment	n8n workflow trials, prompt experiments, model comparisons	None — sandbox only
L2 — Controlled Change	New agent capability, new production workflow	Must pass Architecture Review
L3 — Architecture Change	AI execution boundary change, Hub strategy change, data contract change	Must produce an ADR

Governance Evolution Model

Stage 1 — Founder Governance (Current)
Archi approves + Simple ADR + Agent Contract

Stage 2 — Platform Governance
Named owners + Review cadence + Automated checks

Stage 3 — Intelligence Governance
AI evaluates AI + Policy enforcement + Continuous improvement

Finnhouses should not adopt Enterprise-scale governance on day one. Stage 1 is the only stage this Blueprint recommends adopting now.

Governance Metrics

Metric	Target
Registered AI Agents	100%
Decisions with Evaluation	100%
Production Workflows with Owner	100%
Secrets without Owner	0
Critical Systems without Rollback	0
Unreviewed AI Changes	0

Architecture Decision Log — Master ADR Register

Strategic Decisions

ADR ID	Title	Ch
4.1	Should all agents route through AiGateway?	4
4.4 / 5.3	Shared Intelligence / AI Memory Layer	4/5
5.1	AI Decision & Feedback Data Model	5
6.1 / 2.1	Hub v1/v2 cutover sequencing	2/6
6.2	Hub state schema reconciliation	6
6.3	Automation Governance Model (n8n)	6
6.4	AI Execution Boundary — Decision #1	6
7.1 / 7.2	Data reuse & learning mechanism design	7
8.4	AI agent data-access scoping	8
13.1 / 13.2	Governance role assignment / maturity level	13

No-Regret Actions

ADR ID	Title	Ch
8.1	Standardize Hub v1/v2 auth header convention	8
8.2 (Phase 1)	Credential Blast Radius Reduction	8
8.3	Recurring security posture review	8
9.1	Hub v1 health-monitoring parity	9
9.2	Proactive alerting thresholds	9
9.3	Rollback playbook + Deployment Confidence Principle	9
9.4	Credential-compromise Response runbook	9

Deferred Decisions

Item	Reason Deferred
Ch2 vs Ch5 table-count discrepancy (28 vs 26)	Requires fresh count at implementation time
Gate 3 — Hub Runtime Evolution final choice	Deferred until Track A/B execution produces real data
hub_state JSON normalization (ADR 5.2)	Depends on Cluster B outcome
Dormant table disposition / schema lifecycle metadata	Low urgency, non-blocking

Domain Dictionary

Term	Definition
Intelligence Layer	Shared substrate (memory, decision history, policy) allowing AI agents to reuse each other's output — currently absent
Decision Ledger	Unified log of every AI decision — currently fragmented across 3 tables
Agent Contract	Minimum required definition of a production AI agent: Identity, Purpose, Input, Output, Decision Impact, Evaluation, Owner
Memory Artifact	Logical object representing accumulated agent knowledge, distinct from a database table
Governance Contract	Output of Phase S0 — an agreement defining an agent's identity/access/evaluation prior to implementation
AI Control Plane	Agent Registry + Policy Engine + Decision Ledger + Evaluation Dataset + Memory Layer + Access Control — not the same as AiGateway
AiGateway	Hub v2 component providing model-routing abstraction — one component of, not equal to, an AI Control Plane
Maturity Inversion Pattern	The most architecturally mature capability exists outside the production boundary
Intelligence Loop Maturity Model (L0–L5)	Scale for platform maturity; current state L1–L2
Value Stream	A structurally distinct path from demand to revenue; Finnhouses has 2 (CRM-routed vs. Fix & Flip)
Business Capability (BC-x.x) / Domain (BD-x.x)	Stable IDs distinguishing what the business does from which application implements it
Architecture Decision Required (ADR Candidate)	A flagged conflict with a stated Principle, recorded with reasoning/drivers rather than resolved immediately
Evidence Standard (A/B/C/D/E)	Source-type classification for every claim
Confidence (High/Medium/Low)	Separate from Evidence — how certain a conclusion is
No-Regret Action	Work reducing real risk without depending on any unresolved strategic decision